

Q3. Docker Storage:-

1. Loss of container-layer data after container deletion.
2. Permanent storage using a Docker named volume and bind mount

1. Create a container

```
ayushi2323@LAPTOP-6LV... x + -
ayushi2323@LAPTOP-6LV...:~$ docker run -dit --name storage-test nginx
8f65d13fd5b8563d922247ccf18faceca47d05f847e3a6c21c632c60c2ebacdfd231e
ayushi2323@LAPTOP-6LV...:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
8f65d13fd5b8   nginx    "/docker-entrypoint.."   11 seconds ago Up 10 seconds 80/tcp       storage-
```

2. Create data inside the container

```
ayushi2323@LAPTOP-6LV...:~$ docker exec storage-test sh -c "echo 'This data is stored in the container layer' > /tmp/test.txt"
ayushi2323@LAPTOP-6LV...:~$ docker exec storage-test cat /tmp/test.txt
```

3. Delete the container

```
ayushi2323@LAPTOP-6LV...:~$ docker stop storage-test
storage-test
ayushi2323@LAPTOP-6LV...:~$ docker rm storage-test
storage-test
ayushi2323@LAPTOP-6LV...:~$ docker run -dit --name storage-test nginx
3c26a71ec5650d2c19205eb712a289149a5228d90da37f55a62fbda4c6a70a59
ayushi2323@LAPTOP-6LV...:~$ docker exec storage-test cat /tmp/test.txt
cat: /tmp/test.txt: No such file or directory
```

4. Create a new container

```
ayushi2323@LAPTOP-6LV...:~$ docker run -dit --name storage storage-test nginx
3c26a71ec5650d2c19205eb712a289149a5228d90da75f5a6fbda4c6a70a59
ayushi2323@LAPTOP-6LV...:~$ docker run -dit --name at docker exec storage-test cat /tmp/test.txt
cat: /tmp/test.txt: No such file or directory
```

5. Create a named volume

```
cat: /tmp/test.txt: No such file or directory
ayushi2323@LAPTOP-6LV...:~$ docker volume create myvolume
myvolume
ayushi2323@LAPTOP-6LV...:~$ docker volume ls
DRIVER      VOLUME NAME
local      myvolume
```

6. Attach the volume to a container

```
ayushi2323@LAPTOP-6LV...:~$ docker volume create myvolume
myvolume
ayushi2323@LAPTOP-6LV...:~$ docker volume ls
DRIVER      VOLUME NAME
local      myvolume
ayushi2323@LAPTOP-6LV...:~$ docker run -dit --name volume-test -v myvolume:/data nginx
42c818f56b83369de8bc172da798c81df70304ddcf8a5cd9438deab18c5a4baabc4baa
```

7. Store data in the volume

```
ayushi2323@LAPTOP-6LV...:~$ docker exec volume-test sh -c "echo 'This data is stored permanently in a Docker volume' > /data/test.txt"
ayushi2323@LAPTOP-6LV...:~$ docker exec volume-test cat /data/test.txt
This data is stored permanently in a Docker volume
```

8. Delete the container

```
ayushi2323@LAPTOP-6LV...:~$ docker stop volume-test
volume-test
ayushi2323@LAPTOP-6LV...:~$ docker rm volume-test
volume-test
```

9. Create another container using the same volume

```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker run -dit --name volume-test-2 -v myvolume:/data nginx
c61b7cc1451807edf5d8d2a57af30580de055954027a4da913838a4c7e4a4f51
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec volume-test-2 cat /data/test.txt
This data is stored permanently in a Docker volume
```

10. Create a directory on your computer

```
ayushi2323@LAPTOP-6LVIDNHT:~$ mkdir ~/docker-data
ayushi2323@LAPTOP-6LVIDNHT:~$ ls ~/docker-data
```

11. Create a container with a bind mount

```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker run -dit --name bind-test -v ~/docker-data:/data nginx
a15627623259cee699ac0c3f6beb427a2aef51afddc26f11d7f6fb99a57351dc
```

12. Create data inside the container

```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec bind-test sh -c "echo 'This data is stored using a bind mount' > /data/test.txt"
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec bind-test cat /data/test.txt
This data is stored using a bind mount
ayushi2323@LAPTOP-6LVIDNHT:~$ cat ~/docker-data/test.txt
This data is stored using a bind mount
```

13. Delete the container

```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker stop bind-test
bind-test
ayushi2323@LAPTOP-6LVIDNHT:~$ docker rm bind-test
bind-test
ayushi2323@LAPTOP-6LVIDNHT:~$ cat ~/docker-data/test.txt
This data is stored using a bind mount
ayushi2323@LAPTOP-6LVIDNHT:~$
```